MATH 128a: Homework #13

Instructor: F. Weber

Due by: **Wednesday, December 10, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Romberg Integration

Solve problems 2ad, 12(1), 17(13), 18(14), 19(15) of Section 4.5 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 1

Exercise 4.5.2. Solution.

(a) $\int_{-1}^{1} (\cos x)^2 \, dx$

Using $\cos^2 x = \frac{1+\cos(2x)}{2}$,

$$\int_{-1}^{1} \cos^2 x \, dx = \int_{-1}^{1} \frac{1+\cos(2x)}{2} \, dx = 1 + \frac{\sin 2}{2} \approx 1.4546487134.$$

Romberg table to $R_{3,3}$:

$R_{1,1}$		
0.5838531635		
$R_{2,1}$	$R_{2,2}$	
1.2919265817	1.5279510545	
$R_{3,1}$	$R_{3,2}$	$R_{3,3}$
1.4161144438	1.4575103978	**1.4528143540**

Thus,

$$R_{3,3} \approx 1.4528143540.$$

(d) $\int_{e}^{2e} \frac{1}{x \ln x} \, dx$

Let $u = \ln x$. Then:

$$\int_{e}^{2e} \frac{1}{x \ln x} \, dx = \int_{1}^{1+\ln 2} \frac{1}{u} \, du = \ln(1 + \ln 2) \approx 0.5265890341.$$

Romberg table to $R_{3,3}$:

$$\begin{array}{lll} R_{1,1} & & \\ 0.6476540273 & & \\ R_{2,1} & R_{2,2} & \\ 0.5609964257 & 0.5321105585 & \\ R_{3,1} & R_{3,2} & R_{3,3} \\ 0.5356089690 & 0.5271464835 & \mathbf{0.5268155452} \end{array}$$

Thus,

$$R_{3,3} \approx 0.5268155452.$$

Exercise 4.5.12. Solution.

Below $R_{k,j}$ denotes the Romberg entries with

$$R_{1,1} = T(h), \qquad R_{k,1} = T(h/2^{\,k-1}), \qquad R_{k,j} = R_{k,j-1} + \frac{R_{k,j-1} - R_{k-1,j-1}}{4^{\,j-1} - 1}.$$

We display the analytic value (when convenient) and the Romberg table up to $R_{3,3}$.

(a) $\displaystyle\int_1^{1.5} x^2 \ln x \, dx.$

Integrate by parts with $u = \ln x,\ dv = x^2 dx$. Then

$$\int x^2 \ln x \, dx = \frac{x^3}{3} \ln x - \frac{x^3}{9} + C,$$

so

$$I_a = \left[\frac{x^3}{3} \ln x - \frac{x^3}{9} \right]_1^{1.5} = \frac{27}{24} \ln(1.5) - \frac{27}{72} + \frac{1}{9} = \tfrac{9}{8} \ln(1.5) - \tfrac{19}{72}.$$

Numerically,

$$I_a \approx 0.1922593577.$$

Romberg table (trapezoid with $n = 1, 2, 4$ panels and extrapolations):

$$\begin{array}{lll} R_{1,1} & & \\ 0.2280741233 & & \\ R_{2,1} & R_{2,2} & \\ 0.2012025114 & 0.1922453074 & \\ R_{3,1} & R_{3,2} & R_{3,3} \\ 0.1944944732 & 0.1922584604 & \mathbf{0.1922593373} \end{array}$$

(b) $\displaystyle\int_0^1 x^2 e^{-x} \, dx.$

This content is protected and may not be shared, uploaded, or distributed.

Use repeated integration by parts (or note the antiderivative)

$$\int x^2 e^{-x}\,dx = -e^{-x}(x^2+2x+2)+C,$$

hence

$$I_b = \Big[-e^{-x}(x^2+2x+2)\Big]_0^1 = 2-\frac{5}{e} \approx 0.1606027941.$$

Romberg table:

$R_{1,1}$		
0.1839397206		
$R_{2,1}$	$R_{2,2}$	
0.1677861928	0.1624016835	
$R_{3,1}$	$R_{3,2}$	$R_{3,3}$
0.1624884051	0.1607224759	**0.1606105287**

(c) $\displaystyle\int_0^{0.35} \frac{2}{x^2-4}\,dx.$

Partial fractions: $\dfrac{2}{x^2-4} = \frac{1}{2}\Big(\dfrac{1}{x-2}-\dfrac{1}{x+2}\Big)$. Thus

$$\int \frac{2}{x^2-4}\,dx = \tfrac{1}{2}\ln\left|\frac{x-2}{x+2}\right|+C,$$

and

$$I_c = \frac{1}{2}\ln\frac{1.65}{2.35} \approx -0.1768200201.$$

Romberg table:

$R_{1,1}$		
−0.1777643456		
$R_{2,1}$	$R_{2,2}$	
−0.1770572633	−0.1768215692	
$R_{3,1}$	$R_{3,2}$	$R_{3,3}$
−0.1768794052	−0.1768201191	**−0.1768200225**

(d) $\displaystyle\int_0^{\pi/4} x^2 \sin x\,dx.$

Integrate by parts twice. One convenient closed form is

$$\int_0^{\pi/4} x^2\sin x\,dx = \Big[-x^2\cos x + 2x\sin x + 2\cos x\Big]_0^{\pi/4} - 2$$

which simplifies to

$$I_d = \frac{\sqrt{2}}{2}\left(\frac{\pi}{2}-\frac{\pi^2}{16}\right)+\sqrt{2}-2.$$

This content is protected and may not be shared, uploaded, or distributed.

Numerically,

$$I_d \approx 0.0887552844.$$

Romberg table:

$$\begin{array}{lll} R_{1,1} & & \\ 0.1712870977 & & \\ R_{2,1} & R_{2,2} & \\ 0.1088185261 & 0.0879956690 & \\ R_{3,1} & R_{3,2} & R_{3,3} \\ 0.0937365345 & 0.0887092039 & \mathbf{0.0887567729} \end{array}$$

In each part the Romberg entry $R_{3,3}$ is shown in bold and agrees well with the exact (or high-accuracy) value given above.

Exercise 4.5.17. Solution.
Let $T(h)$ denote the composite trapezoidal rule with step size h. Fix an even refinement so that $h = h_k$ and $h_{k-1} = 2h$. Take $2m$ subintervals for $T(h)$ (so the interval endpoints are $x_0 = a, x_1 = a + h, \ldots, x_{2m} = b$). Then

$$T(h) = \frac{h}{2}\Big(f(x_0) + 2\sum_{j=1}^{2m-1} f(x_j) + f(x_{2m})\Big),$$

while with step $2h$ (that is, m subintervals)

$$T(2h) = \frac{2h}{2}\Big(f(x_0) + 2\sum_{j=1}^{m-1} f(x_{2j}) + f(x_{2m})\Big) = h\Big(f(x_0) + 2\sum_{j=1}^{m-1} f(x_{2j}) + f(x_{2m})\Big).$$

Form the combination $4T(h) - T(2h)$. Using the splitting of the sum $\sum_{j=1}^{2m-1} f(x_j) = \sum_{r=1}^{m} f(x_{2r-1}) + \sum_{r=1}^{m-1} f(x_{2r})$ we get

$$\begin{aligned} 4T(h) - T(2h) &= 2h\Big(f(x_0) + 2\sum_{j=1}^{2m-1} f(x_j) + f(x_{2m})\Big) - h\Big(f(x_0) + 2\sum_{r=1}^{m-1} f(x_{2r}) + f(x_{2m})\Big) \\ &= h\Big(f(x_0) + 4\sum_{r=1}^{m} f(x_{2r-1}) + 2\sum_{r=1}^{m-1} f(x_{2r}) + f(x_{2m})\Big). \end{aligned}$$

Recognize the right-hand side as 3 times the composite Simpson rule with step h:

$$S_{\text{comp}}(h) = \frac{h}{3}\Big(f(x_0) + 4\sum_{r=1}^{m} f(x_{2r-1}) + 2\sum_{r=1}^{m-1} f(x_{2r}) + f(x_{2m})\Big).$$

Hence

$$S_{\text{comp}}(h) = \frac{4T(h) - T(2h)}{3}.$$

This content is protected and may not be shared, uploaded, or distributed.

Finally, Romberg extrapolation gives

$$R_{k,2} = R_{k,1} + \frac{R_{k,1} - R_{k-1,1}}{4-1} = \frac{4R_{k,1} - R_{k-1,1}}{3} = \frac{4T(h) - T(2h)}{3} = S_{\text{comp}}(h).$$

This proves that $R_{k,2}$ equals the composite Simpson's rule with $h = h_k$.

Exercise 4.5.18. Solution.
Note that

$$a + \frac{i}{2}h_{k-1} \quad \text{with} \quad i = 1, \dots, 2^{k-1} - 1$$

produces all points that lie halfway between the nodes of step size h_{k-1}. Split the index i into odd and even values.
If $i = 2j - 1$ (odd), then

$$a + \frac{2j-1}{2}h_{k-1} = a + \left(j - \tfrac{1}{2}\right) h_{k-1}, \qquad j = 1, \dots, 2^{k-2},$$

which yields the first sum on the right-hand side.
If $i = 2j$ (even), then

$$a + \frac{2j}{2}h_{k-1} = a + jh_{k-1}, \qquad j = 1, \dots, 2^{k-2} - 1,$$

which yields the second sum.
Thus the original sum is exactly the union of these two sets of terms, giving

$$\sum_{i=1}^{2^{k-1}-1} f\left(a + \frac{i}{2}h_{k-1}\right) = \sum_{i=1}^{2^{k-2}} f\left(a + \left(i - \tfrac{1}{2}\right) h_{k-1}\right) + \sum_{i=1}^{2^{k-2}-1} f(a + ih_{k-1}),$$

as required.

Exercise 4.5.19. Solution.

Recall that

$$R_{k,1} = T(h_k) \quad \text{and} \quad T(h_k) = \frac{h_k}{2}\left[f(a) + 2\sum_{i=1}^{2^k-1} f(a + ih_k) + f(b)\right],$$

with $h_k = h_{k-1}/2$. Using step size $h_k = \frac{1}{2}h_{k-1}$,

$$a + ih_k = a + \frac{i}{2}h_{k-1}.$$

Thus

$$R_{k,1} = \frac{1}{2}\left[R_{k-1,1} + h_k \sum_{i=1}^{2^k-1} f\left(a + \frac{i}{2}h_{k-1}\right)\right].$$

By Exercise 14,

$$\sum_{i=1}^{2^k-1} f\left(a + \frac{i}{2}h_{k-1}\right) = \sum_{i=1}^{2^{k-2}} f\left(a + \left(i - \tfrac{1}{2}\right) h_{k-1}\right) + \sum_{i=1}^{2^{k-2}-1} f(a + ih_{k-1}).$$

This content is protected and may not be shared, uploaded, or distributed.

Substituting this identity and noting that

$$R_{k-1,1} = \frac{h_{k-1}}{2}\Big[f(a) + 2\sum_{i=1}^{2^{k-2}-1} f(a + ih_{k-1}) + f(b)\Big],$$

the terms involving $\sum_{i=1}^{2^{k-2}-1} f(a + ih_{k-1})$ combine, leaving

$$R_{k,1} = \frac{1}{2}\left[R_{k-1,1} + h_{k-1}\sum_{i=1}^{2^{k-2}} f\left(a + \left(i - \tfrac{1}{2}\right)h_{k-1}\right)\right].$$

This content is protected and may not be shared, uploaded, or distributed.

Problem 2: ODEs

Solve problems 1ad, 4bd, 6a(8a), 10(9) of Section 5.1 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

Answer of Problem 2

Exercise 5.1.1. Solution.
(a) For the IVP

$$y' = y\cos t, \qquad 0 \le t \le 1, \qquad y(0) = 1,$$

the function $f(t, y) = y\cos t$ and its partial derivative

$$\frac{\partial f}{\partial y} = \cos t$$

are continuous on the strip $0 \le t \le 1$. Thus f satisfies a Lipschitz condition in y, and Theorem 5.4 guarantees a unique solution.
The equation is separable:

$$\frac{y'}{y} = \cos t \quad \Rightarrow \quad \ln y = \sin t + C.$$

Using $y(0) = 1$ gives $C = 0$, so the unique solution is

$$\boxed{y(t) = e^{\sin t}}.$$

(d) For the IVP

$$y' = \frac{4t^3}{1+t^4}\, y, \qquad 0 \le t \le 1, \qquad y(0) = 1,$$

the function

$$f(t, y) = \frac{4t^3}{1+t^4}\, y$$

and its partial derivative

$$\frac{\partial f}{\partial y} = \frac{4t^3}{1+t^4}$$

are continuous on the interval, so f is Lipschitz in y. By Theorem 5.4, a unique solution exists.
The equation is separable:

$$\frac{y'}{y} = \frac{4t^3}{1+t^4} \quad \Rightarrow \quad \ln y = \int \frac{4t^3}{1+t^4}\, dt = \ln(1+t^4) + C.$$

Applying $y(0) = 1$ gives $C = 0$, so the unique solution is

$$\boxed{y(t) = 1 + t^4}.$$

Exercise 5.1.4. Solution.
(b) $f(t, y) = \dfrac{1+y}{1+t}$.

This content is protected and may not be shared, uploaded, or distributed.

i. We compute

$$\frac{\partial f}{\partial y} = \frac{1}{1+t}.$$

For $0 \le t \le 1$, this derivative is continuous and bounded, so f satisfies a Lipschitz condition in y on D.

ii. Since f is continuous and Lipschitz in y, Theorem 5.6 applies, and the IVP is well-posed.

(d) $f(t,y) = \dfrac{y^2}{1+t}$.

i.

$$\frac{\partial f}{\partial y} = \frac{2y}{1+t}.$$

This derivative is unbounded as $|y| \to \infty$. Thus f does not satisfy a Lipschitz condition in y on all of D.

ii. Since f is not Lipschitz in y on D, Theorem 5.6 cannot be applied to conclude well-posedness.

Exercise 5.1.6. Solution.

Consider the IVP

$$y' = 1 - y, \qquad 0 \le t \le 2, \qquad y(0) = 0.$$

Define $f(t,y) = 1 - y$. Then

$$\frac{\partial f}{\partial y} = -1,$$

which is continuous and bounded on the entire strip $0 \le t \le 2,\ -\infty < y < \infty$. Hence f satisfies a global Lipschitz condition in y with Lipschitz constant $L = 1$. This implies that small perturbations in the initial data or in the function produce proportionally small changes in the solution, including under perturbations of the form $\delta(t) = \delta t$. Therefore, the IVP is well-posed.
To find the solution, solve the linear ODE:

$$y' + y = 1.$$

The integrating factor is e^t, so

$$\frac{d}{dt}(e^t y) = e^t \quad \Longrightarrow \quad e^t y = e^t + C.$$

Using $y(0) = 0$ gives $C = -1$. Thus

$$y(t) = 1 - e^{-t}.$$

Hence the IVP is well-posed and its solution is

$$y(t) = 1 - e^{-t}, \qquad 0 \le t \le 2.$$

This content is protected and may not be shared, uploaded, or distributed.

Exercise 5.1.10. Solution.
(a) For the IVP

$$y' = f(t,y), \qquad y(a) = \alpha,$$

integrate both sides from a to t:

$$y(t) - y(a) = \int_a^t f(\tau, y(\tau))\, d\tau.$$

Using $y(a) = \alpha$, this becomes

$$y(t) = \alpha + \int_a^t f(\tau, y(\tau))\, d\tau.$$

Picard's method replaces y inside the integral by successive approximations:

$$y_0(t) = \alpha, \qquad y_k(t) = \alpha + \int_a^t f(\tau, y_{k-1}(\tau))\, d\tau, \qquad k \geq 1.$$

(b) Apply Picard iteration to the IVP

$$y' = -y + t + 1, \qquad 0 \leq t \leq 1, \qquad y(0) = 1.$$

Thus $f(t,y) = -y + t + 1$ and $\alpha = 1$.

$$y_0(t) = 1.$$

$$y_1(t) = 1 + \int_0^t \big[-1 + \tau + 1 \big]\, d\tau = 1 + \int_0^t \tau\, d\tau = 1 + \frac{t^2}{2}.$$

$$y_2(t) = 1 + \int_0^t \left[-\left(1 + \frac{\tau^2}{2}\right) + \tau + 1 \right] d\tau = 1 + \int_0^t \left(\tau - \frac{\tau^2}{2}\right) d\tau = 1 + \frac{t^2}{2} - \frac{t^3}{6}.$$

$$y_3(t) = 1 + \int_0^t \left[-\left(1 + \frac{\tau^2}{2} - \frac{\tau^3}{6}\right) + \tau + 1 \right] d\tau = 1 + \int_0^t \left(\tau - \frac{\tau^2}{2} + \frac{\tau^3}{6}\right) d\tau.$$

Thus

$$y_3(t) = 1 + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24}.$$

The Picard iterates are therefore

$$\begin{aligned} y_0(t) &= 1, \\ y_1(t) &= 1 + \frac{t^2}{2}, \\ y_2(t) &= 1 + \frac{t^2}{2} - \frac{t^3}{6}, \\ y_3(t) &= 1 + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24}. \end{aligned}$$

This content is protected and may not be shared, uploaded, or distributed.

(c) The exact solution of the IVP is

$$y(t) = t + e^{-t}.$$

Expand e^{-t} in a Maclaurin series:

$$e^{-t} = 1 - t + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24} - \cdots.$$

Thus

$$y(t) = t + e^{-t} = t + \left(1 - t + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24} - \cdots\right),$$

so

$$y(t) = 1 + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24} - \cdots.$$

Comparing with the Picard iterates:

$$y_1(t) = 1 + \frac{t^2}{2}, \qquad y_2(t) = 1 + \frac{t^2}{2} - \frac{t^3}{6}, \qquad y_3(t) = 1 + \frac{t^2}{2} - \frac{t^3}{6} + \frac{t^4}{24},$$

we see that each Picard iterate adds one more term of the Maclaurin series of the exact solution.

This content is protected and may not be shared, uploaded, or distributed.

Problem 3: Euler's method

Solve problems 2, 4, 10, 12 of Section 5.2 of the textbook.
(If the problem has a different number in the 9th edition book, it is in parentheses)

(For 10c, compute a Lipschitz contant on the region $D = [1,2] \times [-1.05, -0.45]$ and argue that the proof of Theorem 5.9 remains valid for sufficiently small h when D is a rectangle $[a,b] \times [c,d]$ instead of a strip $[a,b] \times \mathbb{R}$ provided that the exact solution and the numerical solution stay within the rectangle. Show that this is the case here.)

Answer of Problem 3

Exercise 5.2.2. Solution.
Euler's Method: $y_{n+1} = y_n + h\, f(t_n, y_n), \qquad t_{n+1} = t_n + h.$
(a) $y' = e^{t-y}, \quad y(0) = 1, \quad h = 0.5.$

$$t_0 = 0, \qquad y_0 = 1.$$

$$y_1 = 1 + 0.5e^{-1} \approx 1.18394.$$

$$y_2 = 1.18394 + 0.5\, e^{0.5 - 1.18394} \approx 1.4406.$$

Thus,

$$y(1) \approx 1.4406.$$

(b) $y' = \dfrac{1+t}{1+y}, \quad y(1) = 2, \quad h = 0.5.$

$$t_0 = 1, \qquad y_0 = 2.$$

$$y_1 = 2 + 0.5\left(\frac{2}{3}\right) = 2.33333.$$

$$y_2 = 2.33333 + 0.5\left(\frac{2.5}{3.33333}\right) \approx 2.70833.$$

Thus,

$$y(2) \approx 2.70833.$$

(c) $y' = -y + t\sqrt{y}, \quad y(2) = 2, \quad h = 0.25.$

$$t_0 = 2, \qquad y_0 = 2.$$

$$y_1 = 2 + 0.25\left[-2 + 2\sqrt{2}\right] \approx 2.2071.$$

$$y_2 = 2.2071 + 0.25\left[-2.2071 + 2.25\sqrt{2.2071}\right] \approx 2.4899.$$

$$y_3 = 2.4899 + 0.25\left[-2.4899 + 2.5\sqrt{2.4899}\right] \approx 2.8510.$$

Thus,

$$y(3) \approx 2.8510.$$

(d) $y' = t^{-2}(\sin(2t) - 2ty), \quad y(1) = 2, \quad h = 0.25.$

$$t_0 = 1, \qquad y_0 = 2.$$

This content is protected and may not be shared, uploaded, or distributed.

$$f(1,2) = \sin 2 - 4 \approx -3.0907, \qquad y_1 = 2 + 0.25(-3.0907) \approx 1.2273.$$
$$f(1.25, 1.2273) \approx -1.5807, \qquad y_2 = 1.2273 + 0.25(-1.5807) \approx 0.8311.$$
$$f(1.5, 0.8311) \approx -1.0454, \qquad y_3 = 0.8311 + 0.25(-1.0454) \approx 0.5697.$$
$$f(1.75, 0.5697) \approx -0.7646, \qquad y_4 = 0.5697 + 0.25(-0.7646) \approx 0.3786.$$

Thus,

$$y(2) \approx 0.3786.$$

Final Euler Approximations:

$$(a)\ y(1) \approx 1.4406,$$
$$(b)\ y(2) \approx 2.7083,$$
$$(c)\ y(3) \approx 2.8510,$$
$$(d)\ y(2) \approx 0.3786.$$

Exercise 5.2.4. Solution.

The exact solutions for the IVPs in Exercise 2 are given. Compute the actual error

$$|y(t_i) - w_i|$$

and compare with the error bound in Theorem 5.9 when applicable.

(a) Exact solution:

$$y(t) = \ln\big(e^t + e - 1\big).$$

From Exercise 2(a), Euler's method with $h = 0.5$ gave

$$w_2 = y(1) \approx 1.4406.$$

Compute the exact value:

$$y(1) = \ln(e + e - 1) = \ln(2e - 1) \approx \ln(4.43656) \approx 1.4913.$$

$$\text{Actual error} = |1.4913 - 1.4406| \approx 0.0507.$$

To apply Theorem 5.9, note $f(t, y) = e^{t-y}$ and

$$\frac{\partial f}{\partial y} = -e^{t-y},$$

so $|\partial f/\partial y| \leq 1$ on the region, hence $L = 1$. Compute $M = \max|y''(t)|$ on $[0, 1]$:

$$y'(t) = e^{t-y(t)}, \qquad y''(t) = e^{t-y}(1 - y').$$

A bound is $M \leq 1$. Thus the error bound is

$$\frac{hM}{2L}(e^{L(1-0)} - 1) = \frac{0.5}{2}(e - 1) \approx 0.5 \times 0.8591 = 0.4296.$$

This content is protected and may not be shared, uploaded, or distributed.

Since

$$0.0507 < 0.4296,$$

the actual error is well within the theoretical bound.

(b) Exact solution:

$$y(t) = \sqrt{t^2 + 2t + 6} - 1.$$

From Exercise 2(b),

$$w_2 = y(2) \approx 2.70833.$$

Exact value:

$$y(2) = \sqrt{4 + 4 + 6} - 1 = \sqrt{14} - 1 \approx 3.74166 - 1 = 2.74166.$$

Actual error:

$$|2.74166 - 2.70833| \approx 0.03333.$$

Check Theorem 5.9:

$$f(t, y) = \frac{1 + t}{1 + y}, \qquad \frac{\partial f}{\partial y} = -\frac{1 + t}{(1 + y)^2}.$$

On the region $1 \le t \le 2,\ 1 \le y \le 3$,

$$|\partial f/\partial y| \le \frac{3}{4} = 0.75,$$

hence $L = 0.75$. A bound for the second derivative yields $M \approx 1$. Thus

$$\frac{hM}{2L}(e^{L(2-1)} - 1) = \frac{0.5}{1.5}(e^{0.75} - 1) \approx 0.3333(1.117 - 1) \approx 0.039.$$

Actual error 0.03333 is within the bound.

(c) Exact solution:

$$y(t) = \left(t - 2 + \sqrt{2e}\, e^{-t/2}\right)^2.$$

From Exercise 2(c),

$$w_3 = y(3) \approx 2.8510.$$

Exact value:

$$y(3) = \left(1 + \sqrt{2e}\, e^{-3/2}\right)^2.$$

Compute:

$$\sqrt{2e} \approx 2.33164, \qquad e^{-3/2} \approx 0.22313,$$
$$\sqrt{2e}\, e^{-3/2} \approx 0.5203,$$
$$y(3) \approx (1 + 0.5203)^2 \approx 2.3133.$$

Actual error:

$$|2.3133 - 2.8510| \approx 0.5377.$$

To apply Theorem 5.9, $f(t, y) = -y + t\sqrt{y}$ does not satisfy a global Lipschitz condition in y on the region (because $\partial f/\partial y$ blows up near $y = 0$), so an error bound from Theorem 5.9 cannot be guaranteed.

This content is protected and may not be shared, uploaded, or distributed.

(d) Exact solution:

$$y(t) = \frac{4 + \cos 2 - \cos 2t}{2t^2}.$$

From Exercise 2(d),

$$w_4 = y(2) \approx 0.3786.$$

Exact value:

$$y(2) = \frac{4 + \cos 2 - \cos 4}{8}.$$

$$\cos 2 \approx -0.4161, \qquad \cos 4 \approx -0.6536,$$

$$y(2) = \frac{4 - 0.4161 + 0.6536}{8} = \frac{4.2375}{8} \approx 0.5297.$$

Actual error:

$$|0.5297 - 0.3786| \approx 0.1511.$$

To apply Theorem 5.9, note

$$f(t, y) = t^{-2}(\sin 2t - 2ty), \qquad \frac{\partial f}{\partial y} = -2t^{-1},$$

which becomes unbounded as $t \to 1$. Thus f is not Lipschitz on the needed rectangle, and Theorem 5.9 does not apply.

Summary of Actual Errors:

$$\begin{aligned} (a): &\quad 0.0507, \\ (b): &\quad 0.0333, \\ (c): &\quad 0.5377, \\ (d): &\quad 0.1511. \end{aligned}$$

In parts (a) and (b), the errors lie below the theoretical bound; in (c) and (d), Theorem 5.9 does not apply.

Exercise 5.2.10. Solution.
Given the IVP

$$y' = \frac{1}{t^2} - \frac{y}{t} - y^2, \qquad 1 \le t \le 2, \qquad y(1) = -1,$$

with exact solution $y(t) = -1/t$.

(a) Euler's method with $h = 0.05$.
Euler's update:

$$y_{n+1} = y_n + h\left(\frac{1}{t_n^2} - \frac{y_n}{t_n} - y_n^2\right), \qquad t_{n+1} = t_n + 0.05.$$

Initial values:

$$t_0 = 1, \qquad y_0 = -1.$$

Step 1:

$$f(1, -1) = 1 - (-1) - 1 = 1, \qquad y_1 = -1 + 0.05(1) = -0.95.$$

This content is protected and may not be shared, uploaded, or distributed.

Step 2:

$$f(1.05, -0.95) = \frac{1}{1.05^2} - \frac{-0.95}{1.05} - (-0.95)^2 \approx 0.9044,$$
$$y_2 = -0.95 + 0.05(0.9044) \approx -0.9048.$$

Continuing this process to $t = 2$ yields:

t	$w(t)$	$y(t) = -1/t$
1.00	−1.0000	−1.0000
1.05	−0.9500	−0.95238
1.10	−0.9048	−0.90909
1.15	−0.8635	−0.86957
1.20	−0.8258	−0.83333
1.25	−0.7915	−0.80000
1.30	−0.7603	−0.76923
1.35	−0.7321	−0.74074
1.40	−0.7066	−0.71429
1.45	−0.6836	−0.68966
1.50	−0.6628	−0.66667
⋮	⋮	⋮
2.00	−0.5105	−0.5000

The Euler approximation is consistently slightly above the exact curve.

(b) Linear interpolation using the computed w_i.
For a point t between t_k and t_{k+1}:

$$w(t) = w_k + \frac{t - t_k}{0.05}(w_{k+1} - w_k).$$

(i) $t = 1.052$ lies between $t_0 = 1.00$ and $t_1 = 1.05$:

$$w(1.052) = -1 + \frac{0.052}{0.05}(-0.95 + 1) = -0.948.$$

Exact:

$$y(1.052) = -\frac{1}{1.052} \approx -0.950.$$

(ii) $t = 1.555$ lies between $t = 1.55$ and $t = 1.60$:

$$w(1.555) \approx -0.6657, \qquad y(1.555) = -\frac{1}{1.555} \approx -0.6430.$$

(iii) $t = 1.978$ lies between $t = 1.95$ and $t = 2.00$:

$$w(1.978) \approx -0.5170, \qquad y(1.978) = -\frac{1}{1.978} \approx -0.5056.$$

(c) Determine h so that

$$|y(t_i) - w_i| \leq 0.05$$

This content is protected and may not be shared, uploaded, or distributed.

using Theorem 5.9.
Here

$$f(t,y) = \frac{1}{t^2} - \frac{y}{t} - y^2.$$

Compute

$$\frac{\partial f}{\partial y} = -\frac{1}{t} - 2y.$$

On the exact solution $y = -1/t$,

$$L = \max\left| -\frac{1}{t} - 2\left(-\frac{1}{t}\right)\right| = \max\left|\frac{1}{t}\right| = 1.$$

Next,

$$y(t) = -\frac{1}{t}, \qquad y'(t) = \frac{1}{t^2}, \qquad y''(t) = -\frac{2}{t^3}.$$

Hence

$$|y''(t)| \le 2 \quad \text{for } 1 \le t \le 2,$$

so $M = 2$.
The error bound from Theorem 5.9:

$$|y(t_i) - w_i| \le \frac{hM}{2L}\left(e^{L(t_i-1)} - 1\right) = h\left(e^{t_i-1} - 1\right).$$

Require this ≤ 0.05 for worst case $t_i = 2$:

$$h\big(e^1 - 1\big) \le 0.05,$$

$$h \le \frac{0.05}{e-1} \approx \frac{0.05}{1.71828} \approx 0.0291.$$

Thus a step size satisfying the error requirement is

$$h \le 0.03.$$

Exercise 5.2.12. Solution.

Consider the IVP

$$y' = -10y, \qquad 0 \le t \le 2, \qquad y(0) = 1,$$

which has exact solution

$$y(t) = e^{-10t}.$$

Euler's method with $h = 0.1$. Euler's update is

$$y_{n+1} = y_n + h(-10y_n) = y_n(1 - 10h) = y_n(1-1) = 0.$$

Thus

$$y_1 = 0, \qquad y_2 = 0, \qquad \dots$$

so Euler's method produces the identically zero solution after the first step.

This content is protected and may not be shared, uploaded, or distributed.

Does this violate Theorem 5.9?
No. Theorem 5.9 provides an error bound that depends on:

1. the Lipschitz constant L of $f(t, y) = -10y$, here $L = 10$;
2. the size of h;
3. an upper bound M on $|y''(t)|$.

The theorem requires h to be sufficiently small—but it does *not* guarantee good behavior for arbitrarily large h.
In this case the stability factor for Euler's method is

$$1 - 10h = 1 - 1 = 0.$$

This places the method exactly on the boundary of instability for the linear test equation $y' = \lambda y$. Thus the numerical method collapses immediately to zero, even though the true solution is not zero.
This behavior is due to Euler's instability for stiff equations with large negative coefficients, not a violation of Theorem 5.9. The theorem still holds; the step size $h = 0.1$ is simply too large for Euler's method to produce a meaningful approximation for this problem.

Conclusion. Euler's method with $h = 0.1$ yields the zero solution after one step. This does *not* violate Theorem 5.9—the step size is too large relative to the stiffness of the equation, so Euler's method becomes numerically unstable.

This content is protected and may not be shared, uploaded, or distributed.

Problem 4: Runge-Kutta methods

Solve problems 2, 6, 14, 29(27), 30, 32 of Section 5.4 of the textbook. You may want to write a matlab function for some of the numerical methods to compute the approximations.
(If the problem has a different number in the 9th edition book, it is in parentheses)

(For 30, the equation should be $y_{i+1} = y_i + ha_1 f(t_i, y_i) + ha_2 f(t_i + h\alpha_2, y_i + h\delta_2 f(t_i, y_i))$ with the notation from the lectures. For simplicity, consider the autonomous case $f(t, y) = g(y)$. It suffices to consider just $f(t, y) = y$, but it is more instructive to leave it in the more general form $f(t, y) = g(y)$.)

Answer of Problem 4

Exercise 5.4.2. 5.4.6. 5.4.14. Solution.

```
function y = rk2(f, t, y0, a2, p1, q1)
% RK2   General second-order Runge-Kutta method
%
%   y = rk2(f, t, y0, a2, p1, q1)
%
%   Implements:
%       k1 = f(tn, yn)
%       k2 = f(tn + p1*h, yn + q1*h*k1)
%       yn+1 = yn + h*((1-a2)*k1 + a2*k2)
%
%   Examples:
%       Modified Euler:  a2=1/2, p1=1,   q1=1
%       Midpoint:        a2=1,   p1=1/2, q1=1/2

h = t(2) - t(1);
N = length(t);
y = zeros(1,N);
y(1) = y0;

for n = 1:N-1
    k1 = f(t(n), y(n));
    k2 = f(t(n) + p1*h, y(n) + q1*h*k1);
    y(n+1) = y(n) + h*((1-a2)*k1 + a2*k2);
end
end


function y = rk4(f, t, y0)
% RK4   Classical 4th-order Runge-Kutta method

h = t(2) - t(1);
N = length(t);
y = zeros(1,N);
```

This content is protected and may not be shared, uploaded, or distributed.

```
y(1) = y0;

for n = 1:N-1
    k1 = f(t(n), y(n));
    k2 = f(t(n)+h/2, y(n)+h*k1/2);
    k3 = f(t(n)+h/2, y(n)+h*k2/2);
    k4 = f(t(n)+h,   y(n)+h*k3);
    y(n+1) = y(n) + h*(k1 + 2*k2 + 2*k3 + k4)/6;
end
end



%% Helper coefficient sets
% Modified Euler (Heun)
ME.a2 = 1/2;   ME.p1 = 1;   ME.q1 = 1;

% Midpoint method
MP.a2 = 1;     MP.p1 = 1/2; MP.q1 = 1/2;


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                           Problem 2(a)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
f = @(t,y) exp(t - y);
y_exact = @(t) log(exp(t) + exp(1) - 1);

h = 0.5;
t = 0:h:1;
y0 = 1;

y_ME  = rk2(f, t, y0, ME.a2, ME.p1, ME.q1);
y_MP  = rk2(f, t, y0, MP.a2, MP.p1, MP.q1);
y_RK4 = rk4(f, t, y0);
y_ex  = y_exact(t);

TableA = table( ...
    t', ...
    y_ME', abs(y_ME'  - y_ex'), ...
    y_MP', abs(y_MP'  - y_ex'), ...
    y_RK4',abs(y_RK4' - y_ex'), ...
    y_ex', ...
    'VariableNames', ...
    {'t','ME','Err_ME','MP','Err_MP','RK4','Err_RK4','Exact'} );

disp('==================== Problem 2(a) ====================')
```

```
disp(TableA)


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                            Problem 2(b)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
f = @(t,y) (1+t)/(1+y);
y_exact = @(t) sqrt(t.^2 + 2*t + 6) - 1;

h = 0.5;
t = 1:h:2;
y0 = 2;

y_ME  = rk2(f, t, y0, ME.a2, ME.p1, ME.q1);
y_MP  = rk2(f, t, y0, MP.a2, MP.p1, MP.q1);
y_RK4 = rk4(f, t, y0);
y_ex  = y_exact(t);

TableB = table( ...
    t', ...
    y_ME', abs(y_ME'  - y_ex'), ...
    y_MP', abs(y_MP'  - y_ex'), ...
    y_RK4',abs(y_RK4' - y_ex'), ...
    y_ex', ...
    'VariableNames', ...
    {'t','ME','Err_ME','MP','Err_MP','RK4','Err_RK4','Exact'} );

disp('==================== Problem 2(b) ====================')
disp(TableB)


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                            Problem 2(c)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
f = @(t,y) -y + t*sqrt(y);
y_exact = @(t) (t - 2 + sqrt(2)*exp(-t/2)).^2;

h = 0.25;
t = 2:h:3;
y0 = 2;

y_ME  = rk2(f, t, y0, ME.a2, ME.p1, ME.q1);
y_MP  = rk2(f, t, y0, MP.a2, MP.p1, MP.q1);
y_RK4 = rk4(f, t, y0);
y_ex  = y_exact(t);

```

This content is protected and may not be shared, uploaded, or distributed.

```
TableC = table( ...
    t', ...
    y_ME', abs(y_ME'  - y_ex'), ...
    y_MP', abs(y_MP'  - y_ex'), ...
    y_RK4',abs(y_RK4' - y_ex'), ...
    y_ex', ...
    'VariableNames', ...
    {'t','ME','Err_ME','MP','Err_MP','RK4','Err_RK4','Exact'} );

disp('==================== Problem 2(c) ====================')
disp(TableC)


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%                          Problem 2(d)
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
f = @(t,y) t.^(-2).*( sin(2*t) - 2*t.*y );
y_exact = @(t) 0.5*t.^(-2).*(4 + cos(2) - cos(2*t));

h = 0.25;
t = 1:h:2;
y0 = 2;

y_ME  = rk2(f, t, y0, ME.a2, ME.p1, ME.q1);
y_MP  = rk2(f, t, y0, MP.a2, MP.p1, MP.q1);
y_RK4 = rk4(f, t, y0);
y_ex  = y_exact(t);

TableD = table( ...
    t', ...
    y_ME', abs(y_ME'  - y_ex'), ...
    y_MP', abs(y_MP'  - y_ex'), ...
    y_RK4',abs(y_RK4' - y_ex'), ...
    y_ex', ...
    'VariableNames', ...
    {'t','ME','Err_ME','MP','Err_MP','RK4','Err_RK4','Exact'} );

disp('==================== Problem 2(d) ====================')
disp(TableD)
```

The output is:

This content is protected and may not be shared, uploaded, or distributed.

```
==================== Problem 2(a) ====================
  t      ME      Err_ME       MP      Err_MP      RK4      Err_RK4      Exact
 ___   ______   _________   ______   _________   ______   __________   ______

  0        1           0        1           0        1            0         1
0.5   1.2181    0.004103   1.2154   0.0014074    1.214   1.7886e-05     1.214
  1   1.4976   0.0076744   1.4932   0.0033589   1.4899    4.116e-05    1.4899

==================== Problem 2(b) ====================
  t      ME      Err_ME       MP      Err_MP      RK4      Err_RK4      Exact
 ___   ______   _________   ______   _________   ______   __________   ______

  1        2           0        2           0        2            0         2
1.5   2.3542    6.47e-05   2.3553   0.0011612   2.3541   1.2618e-06    2.3541
  2   2.7417   8.7696e-05   2.7435   0.0018552   2.7417   1.6988e-06    2.7417

==================== Problem 2(c) ====================
  t      ME      Err_ME       MP      Err_MP      RK4      Err_RK4      Exact
 ___   ______   _________   ______   _________   ______   __________   ______

  2        2           0        2           0        2            0         2
2.25   2.2455      1.7426   2.2446      1.7418   2.2441       1.7413    0.50286
 2.5   2.5672      1.7478   2.5653      1.7459   2.5644       1.7451    0.81935
2.75   2.9692      1.7425   2.9663      1.7396   2.9652       1.7385     1.2267
   3   3.4566      1.7259   3.4527       1.722   3.4513       1.7206     1.7307

==================== Problem 2(d) ====================
  t      ME      Err_ME       MP      Err_MP      RK4      Err_RK4      Exact
 ___   ______   _________   ______   _________   ______   __________   ______

  1        2           0        2           0        2            0         2
1.25   1.4161    0.012876   1.4365    0.033311   1.4034   0.00015765    1.4032
 1.5    1.031    0.014601   1.0517    0.035264   1.0166   0.00014844    1.0164
1.75   0.75227   0.014257   0.76936   0.031348   0.73813  0.00012191    0.73801
   2   0.54325   0.013558   0.55668   0.026992   0.52979  9.8467e-05    0.52969
```

Exercise 5.4.29. Solution.
We compare the Midpoint method and the Modified Euler method for the IVP

$$y' = -y + t + 1, \qquad 0 \le t \le 1, \qquad y(0) = 1.$$

Midpoint method:

$$k_1 = f(t_n, y_n), \qquad k_2 = f\left(t_n + \tfrac{h}{2},\, y_n + \tfrac{h}{2}k_1\right),$$
$$y_{n+1} = y_n + hk_2.$$

Modified Euler (Improved Euler) method:

$$k_1 = f(t_n, y_n), \qquad k_2 = f(t_n + h,\, y_n + hk_1),$$
$$y_{n+1} = y_n + \tfrac{h}{2}(k_1 + k_2).$$

For our differential equation

$$f(t, y) = -y + t + 1,$$

which is affine (linear) in both t and y.

Step 1: Compute the Midpoint predictor.

$$k_1 = -y_n + t_n + 1.$$

This content is protected and may not be shared, uploaded, or distributed.

$$y_n + \frac{h}{2}k_1 = y_n + \frac{h}{2}(-y_n + t_n + 1) = (1 - \tfrac{h}{2})y_n + \tfrac{h}{2}(t_n + 1).$$

$$t_n + \tfrac{h}{2}.$$

Then

$$k_2 = -\left((1 - \tfrac{h}{2})y_n + \tfrac{h}{2}(t_n + 1)\right) + \left(t_n + \tfrac{h}{2}\right) + 1.$$

Simplifying yields

$$k_2 = -y_n + t_n + 1 + \frac{h}{2}(-k_1).$$

Thus the Midpoint update becomes

$$y_{n+1} = y_n + hk_2 = y_n + h\left(k_1 + \frac{h}{2}(-k_1)\right) = y_n + \frac{h}{2}(k_1 + k_2),$$

the same form as Modified Euler.

Step 2: Why do they match?
Because $f(t, y)$ is *linear in* y and *linear in* t, the slope changes at a constant rate. Thus evaluating the slope at the midpoint or averaging the endpoint slopes produces *exactly the same* second-order approximation.

Conclusion. For the differential equation

$$y' = -y + t + 1,$$

the Midpoint method and the Modified Euler method give identical approximations for any step size h, because the function $f(t, y)$ is affine (linear) and therefore both methods compute the same effective slope correction.

Exercise 5.4.30. Solution.
Let $y(t)$ denote the exact solution of the IVP. The Taylor expansion of the true solution is

$$y(t_i + h) = y(t_i) + hf + \frac{h^2}{2}(f_t + f_y f) + O(h^3),$$

where all derivatives are evaluated at $(t_i, y(t_i))$.
To compute the local truncation error, insert $w_i = y(t_i)$ into the method and expand the second function evaluation:

$$f(t_i + \alpha_2 h,\ y_i + \delta_2 hf) = f + \alpha_2 hf_t + \delta_2 hf_y f + O(h^2).$$

Hence the method produces

$$w_{i+1} = y_i + a_1 f + a_2\big(f + \alpha_2 hf_t + \delta_2 hf_y f\big) + O(h^2).$$

Collecting terms,

$$w_{i+1} = y_i + (a_1 + a_2)f + ha_2(\alpha_2 f_t + \delta_2 f_y f) + O(h^2).$$

To obtain local truncation error $O(h^3)$, the numerical method must match the Taylor expansion of the exact solution term by term.

This content is protected and may not be shared, uploaded, or distributed.

Matching the coefficient of hf. From the Taylor expansion, the coefficient of hf must be 1. Thus we would need

$$a_1 + a_2 = 1.$$

But in the numerical update above, the coefficient of f is *not* multiplied by h. That is, the method produces a term of the form

$$(a_1 + a_2)f,$$

while the exact solution has a leading term

$$hf.$$

Since a_1 and a_2 are constants independent of h, there is no way to make

$$a_1 + a_2 = h$$

hold for all sufficiently small h. Thus the leading term of the numerical method cannot match the leading term of the Taylor expansion.

Conclusion. Because the first–order term cannot be matched, the method fails to reproduce even the $O(h)$ term of the Taylor expansion, and therefore cannot achieve local truncation error $O(h^3)$ for any choice of constants $a_1, a_2, \alpha_2, \delta_2$.

$$\boxed{\text{The method cannot have local truncation error of order } O(h^3).}$$

Exercise 5.4.32. Solution.
The classical fourth–order Runge–Kutta method is

$$\begin{aligned} k_1 &= f(t_i, w_i), \\ k_2 &= f\left(t_i + \tfrac{h}{2},\, w_i + \tfrac{h}{2}k_1\right), \\ k_3 &= f\left(t_i + \tfrac{h}{2},\, w_i + \tfrac{h}{2}k_2\right), \\ k_4 &= f\left(t_i + h,\, w_i + hk_3\right), \end{aligned}$$

and

$$w_{i+1} = w_i + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4).$$

The problem rewrites these four stages in the form

$$\begin{aligned} w_{i+1} = w_i &+ \frac{h}{6} f(t_i, w_i) + \frac{h}{3} f(t_i + \alpha_1 h,\ w_i + \delta_1 h f(t_i, w_i)) \\ &+ \frac{h}{3} f(t_i + \alpha_2 h,\ w_i + \delta_2 h f(t_i + \gamma_2 h,\ w_i + \gamma_3 h f(t_i, w_i))) \\ &+ \frac{h}{6} f(t_i + \alpha_3 h,\ w_i + \delta_3 h f(t_i + \gamma_4 h,\ w_i + \gamma_5 h f(t_i + \gamma_6 h,\ w_i + \gamma_7 h f(t_i, w_i)))) . \end{aligned}$$

To determine the constants, we match each stage with the corresponding stage of the standard RK4 scheme.

This content is protected and may not be shared, uploaded, or distributed.

Second stage. RK4 uses

$$t_i + \frac{h}{2}, \qquad w_i + \frac{h}{2}k_1.$$

Thus

$$\alpha_1 = \frac{1}{2}, \qquad \delta_1 = \frac{1}{2}.$$

Third stage. RK4 uses

$$k_2 = f\left(t_i + \frac{h}{2},\, w_i + \frac{h}{2}k_1\right).$$

Hence

$$\gamma_2 = \frac{1}{2}, \qquad \gamma_3 = \frac{1}{2}.$$

The outer evaluation uses

$$t_i + \frac{h}{2}, \qquad w_i + \frac{h}{2}k_2,$$

so

$$\alpha_2 = \frac{1}{2}, \qquad \delta_2 = \frac{1}{2}.$$

Fourth stage. RK4 uses

$$k_3 = f\left(t_i + \frac{h}{2},\, w_i + \frac{h}{2}k_2\right), \qquad k_4 = f(t_i + h,\, w_i + hk_3).$$

Thus the outermost stage requires

$$\alpha_3 = 1, \qquad \delta_3 = 1.$$

Matching the nested evaluations gives

$$\gamma_4 = \frac{1}{2}, \qquad \gamma_5 = \frac{1}{2}, \qquad \gamma_6 = \frac{1}{2}, \qquad \gamma_7 = \frac{1}{2}.$$

Final values of the constants.

$$\boxed{\begin{array}{llll} \alpha_1 = \frac{1}{2}, & \alpha_2 = \frac{1}{2}, & \alpha_3 = 1, & \\ \delta_1 = \frac{1}{2}, & \delta_2 = \frac{1}{2}, & \delta_3 = 1, & \\ \gamma_2 = \frac{1}{2}, & \gamma_3 = \frac{1}{2}, & \gamma_4 = \frac{1}{2}, & \gamma_5 = \frac{1}{2}, \\ \gamma_6 = \frac{1}{2}, & \gamma_7 = \frac{1}{2}. & & \end{array}}$$

These values reproduce exactly the classical fourth–order Runge–Kutta method.

This content is protected and may not be shared, uploaded, or distributed.